Define in 4 sentences

1.What is a computer what are the main components of computer hardware

A computer is a machine that can be programmed to carry out algorithms, using input and output functions. The main components of a computer include the hard drive, the CPU, and RAM. A Hard drive is the storage device for a computer. A CPU is in charge of the logic that the computer has, and RAM is memory that can be called upon quickly to helo a computer function.

2. Explain the compilation process

The compilation process involves taking source code (from the programmer) and converting it into object code. In other words it takes a higher level language and converts it into a lower level language such as bytes which allows the computer to read the code. As it converts this code it scans the source code for errors, as well as skips over unnecessary writing such as comments. This whole process is executed before the computer starts the execution process.

3. What is the difference between an interpreter and a compiler

An interpreter executes the code line by line providing an immediate response as soon as the computer reads the source code. A compiler reads all of the source code and then executes the functions. This means that an interpreter is typically quicker to produce output. However, a compiler must convert the source code into object code consisting of 1's and 0's before it executes the functions.

Define the following terms in 2 sentences

1. High-Level Language: A form of computer script that uses more common phrases to the user. A high level language is also easier to use, as it allows for keywords that give the user access to shortcuts.
2. Machine Learning: This is a process that Artificial Intelligence goes through in order to learn new information. It is meant to mimic how humans learn, allowing for an accurate replacement of human judgement.
3. Software Development: This is the process of thinking of, creating, testing, and fixing code. Software development is a process which programmers go through in order to create new software for computers wether it is local or public.
4. Programming: This is the process of writing code for a computer to execute. This can be done in many different languages but the result is for a computer to process data in some way.
5. Algorithm: This is a process to be followed during the execution of problem solving. This is used for calculations involving parts such as addition and multiplication.
6. Compiler: A compiler is a computer program that takes a higher level language and converts it into a lower level language so it can be executed by a computer. An example of this is Java turning higher level code into bytes.

7. Java Virtual Machine: This is a virtual machine that allows a computer to read and therefore run off of Java byte code. This means that it is able to read the binary code after Java compiles it.

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The default interactive shell is now zsh.
To update your account to use zsh, please run `chsh -s /bin/zsh`.
For more details, please visit https://support.apple.com/kb/HT208050.
Erics-MBP:~ ericemmendorfer$ java -version
java version "16.0.2" 2021-07-20
Java(TM) SE Runtime Environment (build 16.0.2+7-67)
Java HotSpot(TM) 64-Bit Server VM (build 16.0.2+7-67, mixed mode, sharing)
Erics-MBP:~ ericemmendorfer$
```